

REFERENCES FOR TRUTH TABLES

Rules for Determining the Truth-Values of Truth-Functional Compounds

Ψ	X	$\sim\Psi$	$(\Psi\rightarrow X)$	$(\Psi\wedge X)$	$(\Psi\vee X)$	$(\Psi\leftrightarrow X)$
T	T	F	T	T	T	T
T	F	F	F	F	T	F
F	T	T	T	F	T	F
F	F	T	T	F	F	T

A formula is a *tautology* if it is true in every row of its truth table.

An argument is *tautologically valid* if in every row of its truth table in which all of the premises are true, the conclusion is also true.

Therefore, an argument with no premises is tautologically valid if its conclusion is a tautology.

OUTLINE OF A PROOF THAT EVERY TAUTOLOGY IS DERIVABLE

[This material is not part of Philosophy 31. It is optional.]

Let the *Characteristic formula of a cell* in a truth table be as follows:

- 1) If the cell contains "T", the characteristic formula is the formula that heads the column.
- 2) If the cell contains "F", the characteristic formula is the *negation* of the formula that heads the column.

Let the *Characteristic formula of a row* in a truth table be the conjunction of the characteristic formulas of the initial cells.

Rules for Determining the Truth-Values of Truth-Functional Compounds
showing the CHARACTERISTIC FORMULA of each CELL.

Ψ	X	$\sim\Psi$	$(\Psi\rightarrow X)$	$(\Psi\wedge X)$	$(\Psi\vee X)$	$(\Psi\leftrightarrow X)$
T	T	F	T	T	T	T
Ψ	X	$\sim\Psi$	$(\Psi\rightarrow X)$	$(\Psi\wedge X)$	$(\Psi\vee X)$	$(\Psi\leftrightarrow X)$
T	F	F	F	F	T	F
Ψ	$\sim X$	$\sim\Psi$	$\sim(\Psi\rightarrow X)$	$\sim(\Psi\wedge X)$	$(\Psi\vee X)$	$\sim(\Psi\leftrightarrow X)$
F	T	T	T	F	T	F
$\sim\Psi$	X	$\sim\Psi$	$(\Psi\rightarrow X)$	$\sim(\Psi\wedge X)$	$(\Psi\vee X)$	$\sim(\Psi\leftrightarrow X)$
F	F	T	T	F	F	T
$\sim\Psi$	$\sim X$	$\sim\Psi$	$(\Psi\rightarrow X)$	$\sim(\Psi\wedge X)$	$(\Psi\vee X)$	$(\Psi\leftrightarrow X)$

Note that in each case, one can derive each of the conditionals whose antecedent is the Characteristic formula of a ROW and whose consequent is the Characteristic formula of one of the cells in that row. For example, regarding the second row (T F), the following forms are all derivable:

- $(\Psi\wedge\sim X)\rightarrow\sim\sim\Psi$
- $(\Psi\wedge\sim X)\rightarrow\sim(\Psi\rightarrow X)$
- $(\Psi\wedge\sim X)\rightarrow\sim(\Psi\wedge X)$
- $(\Psi\wedge\sim X)\rightarrow(\Psi\vee X)$
- $(\Psi\wedge\sim X)\rightarrow\sim(\Psi\leftrightarrow X)$

[Optional homework problem: Prove that each of the above, and their analogues for the three other rows in the truth table, are theorems. That is, give proofs for each of the following and their analogs for the other three rows:

$(P \wedge \sim Q) \rightarrow \sim \sim P$
 $(P \wedge \sim Q) \rightarrow \sim (P \rightarrow Q)$
 $(P \wedge \sim Q) \rightarrow \sim (P \wedge Q)$
 $(P \wedge \sim Q) \rightarrow (P \vee Q)$
 $(P \wedge \sim Q) \rightarrow \sim (P \leftrightarrow Q)$]

Thus, in any given row of a truth-table, we can derive the Characteristic formula of any compound from the Characteristic formulas of its components.

Therefore (using induction on the structure of the compound) we can ultimately derive the Characteristic formula of any compound in any row from the Characteristic formula of the row.

Now a tautology Φ is T in every row of its truth-table, so Φ is its own Characteristic formula in every row.

Hence, if Φ is a tautology, and Γ is Characteristic formula of *any* row of its truth-table, we can ultimately derive:

$\Gamma \rightarrow \Phi$

If there were only one sentence letter, "P", in Φ , there would be exactly two rows, whose Characteristic formulas would be "P" and " $\sim P$ ". Hence, the preceding result implies that we could derive both:

$P \rightarrow \Phi$

$\sim P \rightarrow \Phi$

from which, by the derived rule Separation of Cases, we can derive:

$(P \vee \sim P) \rightarrow \Phi$

Now, since we can separately derive:

$P \vee \sim P$

We can plainly derive:

Φ

This completes the argument for the case of tautologies that contain only one sentence letter.

When there are n sentence letters in Φ , there will be 2^n rows in the truth-table, and again, since we can derive each of the following:

$$\Gamma_1 \rightarrow \Phi$$

$$\Gamma_2 \rightarrow \Phi$$

$$\Gamma_3 \rightarrow \Phi$$

...

$$\Gamma_{2^n} \rightarrow \Phi$$

we can, as above, use the derived rule Separation of Cases to derive:

$$(\Gamma_1 \vee \Gamma_2 \vee \Gamma_3 \vee \dots \vee \Gamma_{2^n}) \rightarrow \Phi$$

It only remains to show that we can always derive the disjunction, $(\Gamma_1 \vee \Gamma_2 \vee \Gamma_3 \vee \dots \vee \Gamma_{2^n})$, of the Characteristic formulas of *all* of the rows in *any* truth-table.

For a truth-table involving two sentence letters, we must be able to derive:

$$(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q) \vee (\sim P \wedge \sim Q)$$

This done using two instances of T59 (" $P \vee \sim P$ " and " $Q \vee \sim Q$ ") and T117 (the basis of the derived rule DIST). [Try it!]

We can now derive the disjunction of the Characteristic formulas of all of the rows for a truth-table involving *three* sentence letters by starting with the preceding,

$$((P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q) \vee (\sim P \wedge \sim Q))$$

conjoining another instance of T59 (" $R \vee \sim R$ ")

$$((P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q) \vee (\sim P \wedge \sim Q)) \wedge (R \vee \sim R)$$

and applying the derived rule DIST several times.

In this way, we can ultimately derive the disjunction,

$$(\Gamma_1 \vee \Gamma_2 \vee \Gamma_3 \vee \dots \vee \Gamma_{2^n})$$

of the Characteristic formulas of all of the rows of any truth-table.

And so, since we can derive

$$(\Gamma_1 \vee \Gamma_2 \vee \Gamma_3 \vee \dots \vee \Gamma_{2^n}) \rightarrow \Phi$$

we can finally derive the tautology

$$\Phi$$